Distributable Code for Visual Studio 2022

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> **Note**
>
> In the lists below...
>
> - [arch] represents the processor architecture identifier, for instance "x86", "x64", or "arm".
> - [locale] represents a specific language, locale, or culture identifier, for instance "ENU", "en-us", or "1033".
> - [version] represents a folder name that uses a version number.
> - [VisualStudioFolder] represents the install location for Visual Studio 2022.

List of Utilities and Build Tools for Visual Studio 2022

This is the "Utilities List" that is referenced in the "Utilities" section of Microsoft Software License Terms for certain editions of Visual Studio 2022 (the "software"). Depending on the specific edition of the software, the software you received may not include all of the files on this list. To determine your rights with respect to the following files, please refer to the Visual Studio License Terms that came with your edition of the software. You may not modify these files. Some Utilities by their nature may make it possible for others to access the devices on which the Utilities are installed. You should delete all Utilities you have installed after you finish debugging or deploying your applications and databases

Diagnostic Tools

IntelliTrace Standalone Collector for Visual Studio 2022

- [VisualStudioFolder]\Common7\IDE\CommonExtensions\Microsoft\IntelliTrace\IntelliTraceCollection.cab

Remote Tools for Visual Studio 2022

- vs_remotetools.exe (x86, x64, ARM64 versions)

Performance Tools for Visual Studio 2022

- [VisualStudioFolder]\Team Tools\Performance Tools\Setups\vs_profiler\[arch]_x64_[locale].exe

Visual C++ Utilities

The "Utilities List" includes all files within the subfolders of the directories specified (if present):

- [VisualStudioFolder]\VC\Redist\MSVC\[version]\debug_nonredist[arch]\
  - Microsoft.VC[version].DebugCRT
  - Microsoft.VC[version].DebugCXXAMP
  - Microsoft.VC[version].DebugMFC
  - Microsoft.VC[version].DebugOpenMP
  - Microsoft.VC[version].OpenMP.LLVM

The "Utilities List" includes the following files within the subfolders of the directories specified:

- [VisualStudioFolder]\VC\Auxiliary\VS\redist\GraphicsDbgRedist\
  - VsGraphicsHelper.dll
  - VsGraphicsResources.dll
- [VisualStudioFolder]\VC\Tools\MSVC\[version]\bin\
  - pgort140.dll
  - pgort140ui.dll
  - pgosweep.exe

Distributable Code Files for Visual Studio 2022

The following section is the "Distributable List" or "REDIST.txt" that is referenced in the "Distributable Code" section of the Microsoft Software License Terms for Visual Studio Enterprise

2022, Visual Studio Professional 2022, Visual Studio Community 2022 ("the software"). If you have a validly licensed copy of such software, you may copy and distribute with your program the unmodified form of the files listed below, subject to the License Terms for the software.

ASP.NET Libraries

The following software components are licensed and supported separately under the Microsoft .NET Library terms located at https://aka.ms/dotnet-library-license-vs . If you do not agree to the license terms for these software components, you may not use them.

- MVC
- Web API
- Web Pages with Razor
- Entity Framework
- SignalR
- Katana
- Microsoft XML Document Transformation

Microsoft Azure

Source

- MobileServices.js
- MobileServices.min.js

Object Code

- Microsoft.WindowsAzure.Mobile.dll
- Microsoft.WindowsAzure.Mobile.resources.dll
- Microsoft.WindowsAzure.Mobile.UI.dll
- Microsoft.WindowsAzure.Ext.dll

Blend and XAML Designers for Visual Studio

Redistributable files for Blend Project and Item Templates for Visual Studio are installed in the following locations:

- [VisualStudioFolder]\Common7\IDE\ProjectTemplates

- [VisualStudioFolder]\Common7\IDE\ItemTemplates
- [VisualStudioFolder]\DesignTools\AppThemes
- [Program Files (x86)]\MSBuild\Microsoft\Expression\Blend\.NETFramework

Blend for Visual Studio

Redistributable files for Blend for Visual Studio are installed in the following locations:

- [Program Files (x86)]\Microsoft SDKs\Expression\Blend\.NETFramework\v4.0
- [Program Files (x86)]\Microsoft SDKs\Expression\Blend\.NETFramework\v4.5

Sample Data Resources

- [VisualStudioFolder]\DesignTools\SampleData

.NET Framework (4.6.2 and later versions)

Subject to the License Terms for the software, you may copy and distribute the following files, unmodified, with your program:

Offline Installer

- dotNetFx-x86-x64-AllOS-ENU.exe (.NET Framework redistributable as present in Visual Studio)
- NDP[ver]-KB[number]-x86-x64-AllOS-ENU.exe (.NET Framework redistributable as present on other channels, such as the Microsoft Download Center) Note: Both files are identical but may use different names for different distribution channels.

Language Packs

- dotNetFx-x86-x64-AllOS-[locale].exe
- NDP[ver]-KB[number]-x86-x64-AllOS-[locale].exe

> **Note**
>
> - Both files are identical but may use different names for different distribution channels.
> - [ver] represents a 3 digit version such 462 or 481. For instance, NDP462-KB[number]-x86-x64-AllOS-[locale].exe

- [number] represents a numeric KB number such as 3151800. For instance, NDP462-KB3151800-x86-x64-AllOS-[locale].exe
- [locale] represents the specific three-letter language identifier. For instance, NDP462-KB3151800-x86-x64-AllOS-DEU.exe
- Language Packs are available for the following (listed here with their associated identifier code): Arabic (ARA), Chinese-Taiwan (CHT), Czech (CSY), Danish (DAN), German (DEU), Greek (ELL), Finnish (FIN), French (FRA), Hebrew (HEB), Hungarian (HUN), Italian (ITA), Japanese (JPN), Korean (KOR), Dutch-Netherlands (NLD), Norwegian (NOR), Polish (PLK), Portuguese-Brazil (PTB), Russian (RUS), Swedish (SVE), Turkish (TRK), Chinese (CHS), Portuguese-Portugal (PTG), Spanish (ESN)

F# Runtime

- Fsharp.Core.dll

ADO.NET

Subject to the License Terms for the software, you may copy and distribute the following files, unmodified, with your program:

- System.Data.dll
- System.Data.DatasetExtensions.dll
- System.Data.OracleClient.dll
- Adonetdiag.dll

DIA SDK

Subject to the License Terms for the software, you may copy and distribute the following files, unmodified, with your program:

- [VisualStudioFolder]\DIA SDK\bin\msdia140.dll
- [VisualStudioFolder]\DIA SDK\bin\amd64\msdia140.dll
- [VisualStudioFolder]\DIA SDK\bin\arm64\msdia140.dll
- [VisualStudioFolder]\DIA SDK\bin\arm\msdia140.dll

Visual C++ Runtime Files

Subject to the License Terms for the software, you may copy and distribute the following files, unmodified, as a part of the installation package of your program:

- [VisualStudioFolder]VC\Redist\MSVC\[version]\MergeModules\Microsoft_VC143_CRT_[arch].msm
- [VisualStudioFolder]VC\Redist\MSVC\[version]\MergeModules\Microsoft_VC143_CXXAMP_[arch].msm
- [VisualStudioFolder]VC\Redist\MSVC\[version]\MergeModules\Microsoft_VC143_MFCLOC_[arch].msm
- [VisualStudioFolder]VC\Redist\MSVC\[version]\MergeModules\Microsoft_VC143_MFC_[arch].msm
- [VisualStudioFolder]VC\Redist\MSVC\[version]\MergeModules\Microsoft_VC143_OpenMP_[arch].msm

Subject to the License Terms for the software, you may copy and distribute with your program any of the files within the following folder and its subfolders except as noted below. You may not modify these files.

- [VisualStudioFolder]\VC\redist

The same Visual C++ Runtime Redistributable package is available from Latest supported Visual C++ Redistributable downloads and Download Visual Studio Tools . Distribution of the Visual C++ Runtime Redistributable package, merge modules, and individual binaries obtained from those pages is limited to licensed Visual Studio users and is subject to its license terms.

> **Note**
>
> You may *not* distribute the contents of the following folders (please see **List of Utilities and Build Tools for Visual Studio 2022** for allowed usage for development/testing purposes):
>
> - [VisualStudioFolder]VC\Redist\MSVC\[version]\debug_nonredist
> - [VisualStudioFolder]\VC\Redist\MSVC\[version]\onecore\debug_nonredist

Subject to the License Terms for the software, you may copy and distribute the following files with your program in your program's application local folder or by deploying them into the Global Assembly Cache (GAC):

- [VisualStudioFolder]\VC\Tools\MSVC\[version]\atlmfc\lib\[arch]\mfcmifc80.dll

Universal Windows Apps and Windows Store Apps

Side-loading of Universal Windows Apps

The AppX files contained in the following locations may be distributed unmodified with your Universal Windows apps that you intend to side-load:

- [Program Files (x86)]\Microsoft SDKs\Windows Kits\10\ExtensionSDKs\Microsoft.VCLibs\14.0\Appx\Retail\[arch]\Microsoft.VCLibs.[arch].14.00.appx
- [Program Files (x86)]\Microsoft SDKs\Windows Kits\10\ExtensionSDKs\Microsoft.VCLibs.Desktop\14.0\Appx\Retail\[arch]\Microsoft.VCLibs.[arch].14.00.Desktop.appx
- [Program Files (x86)]\Microsoft SDKs\Windows Kits\10\ExtensionSDKs\Microsoft.VCLibs.120\14.0\Appx\Retail\[arch]\Microsoft.VCLibs.[arch].12.00.Universal.appx
- [Program Files (x86)]\Microsoft SDKs\Windows Kits\10\ExtensionSDKs\Microsoft.NET.Native.Framework.1.3\1.3\[arch]\ret\Native\Microsoft.NET.Native.Framework.1.3.appx
- [Program Files (x86)]\Microsoft SDKs\Windows Kits\10\ExtensionSDKs\Microsoft.NET.Native.Runtime.1.4\1.4\AppX\[arch]\Microsoft.NET.Native.Runtime.1.4.appx
- For additional versions of .NET Native, see https://www.nuget.org/packages/Microsoft.Net.Native.Compiler/ .

The files contained in the following locations may be distributed unmodified with your Universal Windows apps that you intend to side-load:

- [Program Files (x86)]\Microsoft SDKs\Windows Kits\10\ExtensionSDKs\CppUnitTestFramework.Universal\15.0\Redist\CommonConfiguration
- [Program Files (x86)]\Microsoft SDKs\Windows Kits\10\ExtensionSDKs\MSTestFramework.Universal\15.0\Redist\CommonConfiguration
- [Program Files (x86)]\Microsoft SDKs\Windows Kits\10\ExtensionSDKs\TestPlatform.Universal\15.0\Redist\CommonConfiguration

SQL Server Database Tooling files

Subject to the License Terms for the software, you may copy and distribute the .dll files and .exe files, unmodified, in this folder with your program:

- [VisualStudioFolder]Common7\IDE\Extensions\Microsoft\SQLDB\DAC\150

SQL Server Redistributable Components

Subject to the License Terms for the software, you may copy and distribute the following files, unmodified, with your program:

- MsSqlCmdLnUtils.msi
- sqlncli.msi
- sqllocaldb.msi
- SQLSysClrTypes.msi

Microsoft WCF Data Services files

Subject to the License Terms for the software, you may copy and distribute the following files, unmodified, with your program:

- Microsoft.Data.Services.dll
- Microsoft.Data.Services.Client.dll
- Microsoft.Data.OData.dll
- Microsoft.Data.Edm.dll
- System.Spatial.dll

Microsoft Visual Studio Tools for Office

Subject to the License Terms for the software, you may copy and distribute the following files, unmodified, with your program:

- Microsoft.Office.Tools.Common.v4.0.Utilities.dll
- Microsoft.Office.Tools.Excel.v4.0.Utilities.dll
- Microsoft.Office.Tools.Outlook.v4.0.Utilities.dll
- Microsoft.Office.Tools.Word.v4.0.Utilities.dll

Subject to the License Terms for the software, you may copy and distribute the following files with your program:

- setup.exe (bootstrapper used to install Office Add-ins)

Distributable Code Files for the Concurrency Visualizer Software Development Kit

Subject to the License Terms for the software, you may copy and distribute the following files, unmodified, with your program:

- Microsoft.ConcurrencyVisualizer.Markers.dll (for .NET 3.5)
- Microsoft.ConcurrencyVisualizer.Markers.dll (for .NET 4.0)
- cvmarkers.h
- cvmarkersobj.h

Distributable Code Files for Visual Studio extension development

This is the "REDIST list" that is referenced in the "Distributable Code" section of the Microsoft Software License Terms for Visual Studio 2022 ("the software"). If you have a validly licensed copy of the software, you may copy and distribute the unmodified object code form of the files listed below, subject to the License Terms for the software.

- [VisualStudioFolder]\VSSDK\VisualStudioIntegration\Redistributables\VS170_piaredist.exe
- [VisualStudioFolder]\VSSDK\VisualStudioIntegration\Redistributables\VSSDKTestHost.exe

Distributable Code Files for Mobile Development with Xamarin

Subject to the License Terms for the software, you may copy and distribute with your application built using Visual Studio or Visual Studio for Mac the object code form of the following files (and associated debug symbol files) as installed within subfolders of the following directories:

On macOS® operating system

- /Library/Frameworks/Xamarin.iOS.framework

- /Library/Frameworks/Xamarin.Android.framework
- /Library/Frameworks/Xamarin.Mac.framework

On Windows operating system:

- [VisualStudioFolder]\MSBuild\Xamarin,
- [VisualStudioFolder]\Common7\IDE\ReferenceAssemblies\Microsoft\Framework\MonoAndroid,
- [VisualStudioFolder]\Common7\IDE\ReferenceAssemblies\Microsoft\Framework\MonoTouch,
- [VisualStudioFolder]\Common7\IDE\ReferenceAssemblies\Microsoft\Framework\Xamarin.iOS,
- [VisualStudioFolder]\Common7\IDE\ReferenceAssemblies\Microsoft\Framework\Xamarin.Mac,
- [VisualStudioFolder]\Common7\IDE\ReferenceAssemblies\Microsoft\Framework\Xamarin.TVOS, or
- [VisualStudioFolder]\Common7\IDE\ReferenceAssemblies\Microsoft\Framework\Xamarin.WatchOS

Xamarin Distributable Code Files

- FSharp.Compiler.CodeDom.dll
- FSharp.Core.dll
- FSharp.Core.optdata
- FSharp.Core.sigdata
- FSharp.Core.xml
- I18N.CJK.dll
- I18N.dll
- I18N.MidEast.dll
- I18N.Other.dll
- I18N.Rare.dll
- I18N.West.dll
- Info.plist
- Ionic.Zip.dll
- Irony.dll
- Java.Interop.dll

- Java.Interop.Tools.Cecil.dll
- Java.Interop.Tools.Diagnostics.dll
- Java.Interop.Tools.JavaCallableWrappers.dll
- libapp.a
- libextension.a
- libmono-2.0.a
- libmono-2.0.dylib
- libmono-android.debug.d.dylib
- libmono-android.debug.d.so
- libmono-android.debug.dylib
- libmono-android.debug.so
- libmono-android.release.d.dylib
- libmono-android.release.d.so
- libmono-android.release.dylib
- libmono-android.release.so
- libmono-btls-shared.d.so
- libmono-btls-shared.so
- libMonoPosixHelper.d.dylib
- libMonoPosixHelper.d.so
- libMonoPosixHelper.dylib
- libMonoPosixHelper.so
- libmono-profiler-log.a
- libmono-profiler-log.d.dylib
- libmono-profiler-log.d.so
- libmono-profiler-log.dylib
- libmono-profiler-log.so
- libmonosgen-2.0.a
- libmonosgen-2.0.d.dylib
- libmonosgen-2.0.d.so
- libmonosgen-2.0.dylib
- libmonosgen-2.0.so
- libtvextension.a
- libwatchextension.a
- libxamarin.a
- libxamarin.dylib

- libxamarin-debug.a
- libxamarin-debug.dylib
- libxammac.a
- libxammac.dylib
- libxammac-debug.a
- libxammac-debug.dylib
- libxammac-system.a
- libxammac-system-debug.a
- libzip.3.0.dylib
- libZipSharp.dll
- libZipSharp.dll.config
- machine.config
- Microsoft.CSharp.dll
- Microsoft.Win32.Primitives.dll
- Microsoft.Win32.Registry.AccessControl.dll
- Microsoft.Win32.Registry.dll
- Mono
- mono.android.dex
- Mono.Android.dll
- Mono.Android.Export.dll
- mono.android.jar
- Mono.Btls.Interface.dll
- Mono.CompilerServices.SymbolWriter.dll
- Mono.CSharp.dll
- Mono.Data.Sqlite.dll
- Mono.Data.Sqlite.dll.config
- Mono.Data.Tds.dll
- Mono.Messaging.dll
- Mono.Posix.dll
- Mono.Security.dll
- MonoTouch.Dialog-1.dll
- monotouch.dll
- MonoTouch.NUnitLite.dll
- monotouch-fixes.dylib
- mscorlib.dll

- netstandard.dll
- OpenTK.dll
- OpenTK.dll.config
- OpenTK-1.0.dll
- OpenTK-1.0.dll.config
- System.AppContext.dll
- System.Collections.Concurrent.dll
- System.Collections.dll
- System.Collections.NonGeneric.dll
- System.Collections.Specialized.dll
- System.ComponentModel.Annotations.dll
- System.ComponentModel.Composition.dll
- System.ComponentModel.DataAnnotations.dll
- System.ComponentModel.dll
- System.ComponentModel.EventBasedAsync.dll
- System.ComponentModel.Primitives.dll
- System.ComponentModel.TypeConverter.dll
- System.config
- System.Configuration.dll
- System.Configuration.Install.dll
- System.Console.dll
- System.Core.dll
- System.Data.Common.dll
- System.Data.dll
- System.Data.Linq.dll
- System.Data.Services.Client.dll
- System.Data.SqlClient.dll
- System.Diagnostics.Contracts.dll
- System.Diagnostics.Debug.dll
- System.Diagnostics.FileVersionInfo.dll
- System.Diagnostics.Process.dll
- System.Diagnostics.StackTrace.dll
- System.Diagnostics.TextWriterTraceListener.dll
- System.Diagnostics.Tools.dll
- System.Diagnostics.TraceEvent.dll

- System.Diagnostics.TraceSource.dll
- System.Diagnostics.Tracing.dll
- System.dll
- System.Drawing.Primitives.dll
- System.Dynamic.Runtime.dll
- System.EnterpriseServices.dll
- System.Globalization.Calendars.dll
- System.Globalization.dll
- System.Globalization.Extensions.dll
- System.IdentityModel.dll
- System.IdentityModel.Selectors.dll
- System.IO.Compression.dll
- System.IO.Compression.FileSystem.dll
- System.IO.Compression.ZipFile.dll
- System.IO.dll
- System.IO.FileSystem.AccessControl.dll
- System.IO.FileSystem.dll
- System.IO.FileSystem.DriveInfo.dll
- System.IO.FileSystem.Primitives.dll
- System.IO.FileSystem.Watcher.dll
- System.IO.IsolatedStorage.dll
- System.IO.MemoryMappedFiles.dll
- System.IO.Pipes.dll
- System.IO.UnmanagedMemoryStream.dll
- System.Json.dll
- System.Linq.dll
- System.Linq.Expressions.dll
- System.Linq.Parallel.dll
- System.Linq.Queryable.dll
- System.Messaging.dll
- System.Net.AuthenticationManager.dll
- System.Net.Cache.dll
- System.Net.dll
- System.Net.Http.dll
- System.Net.Http.WinHttpHandler.dll

- System.Net.HttpListener.dll
- System.Net.Mail.dll
- System.Net.NameResolution.dll
- System.Net.NetworkInformation.dll
- System.Net.Ping.dll
- System.Net.Primitives.dll
- System.Net.Requests.dll
- System.Net.Security.dll
- System.Net.ServicePoint.dll
- System.Net.Sockets.dll
- System.Net.Utilities.dll
- System.Net.WebHeaderCollection.dll
- System.Net.WebSockets.Client.dll
- System.Net.WebSockets.dll
- System.Numerics.dll
- System.Numerics.Vectors.dll
- System.ObjectModel.dll
- System.Reflection.Context.dll
- System.Reflection.DispatchProxy.dll
- System.Reflection.dll
- System.Reflection.Emit.dll
- System.Reflection.Emit.ILGeneration.dll
- System.Reflection.Emit.Lightweight.dll
- System.Reflection.Extensions.dll
- System.Reflection.Primitives.dll
- System.Reflection.TypeExtensions.dll
- System.Resources.ReaderWriter.dll
- System.Resources.ResourceManager.dll
- System.Runtime.CompilerServices.VisualC.dll
- System.Runtime.dll
- System.Runtime.Extensions.dll
- System.Runtime.Handles.dll
- System.Runtime.InteropServices.dll
- System.Runtime.InteropServices.RuntimeInformation.dll
- System.Runtime.InteropServices.WindowsRuntime.dll

- System.Runtime.Numerics.dll
- System.Runtime.Serialization.dll
- System.Runtime.Serialization.Formatters.dll
- System.Runtime.Serialization.Formatters.Soap.dll
- System.Runtime.Serialization.Json.dll
- System.Runtime.Serialization.Primitives.dll
- System.Runtime.Serialization.Xml.dll
- System.Security.AccessControl.dll
- System.Security.Claims.dll
- System.Security.Cryptography.Algorithms.dll
- System.Security.Cryptography.Cng.dll
- System.Security.Cryptography.Csp.dll
- System.Security.Cryptography.DeriveBytes.dll
- System.Security.Cryptography.Encoding.dll
- System.Security.Cryptography.Encryption.Aes.dll
- System.Security.Cryptography.Encryption.dll
- System.Security.Cryptography.Encryption.ECDiffieHellman.dll
- System.Security.Cryptography.Encryption.ECDsa.dll
- System.Security.Cryptography.Hashing.Algorithms.dll
- System.Security.Cryptography.Hashing.dll
- System.Security.Cryptography.OpenSsl.dll
- System.Security.Cryptography.Pkcs.dll
- System.Security.Cryptography.Primitives.dll
- System.Security.Cryptography.ProtectedData.dll
- System.Security.Cryptography.RandomNumberGenerator.dll
- System.Security.Cryptography.RSA.dll
- System.Security.Cryptography.X509Certificates.dll
- System.Security.dll
- System.Security.Principal.dll
- System.Security.Principal.Windows.dll
- System.Security.SecureString.dll
- System.ServiceModel.dll
- System.ServiceModel.Duplex.dll
- System.ServiceModel.Http.dll
- System.ServiceModel.Internals.dll

- System.ServiceModel.NetTcp.dll
- System.ServiceModel.Primitives.dll
- System.ServiceModel.Security.dll
- System.ServiceModel.Web.dll
- System.ServiceProcess.ServiceController.dll
- System.Text.Encoding.CodePages.dll
- System.Text.Encoding.dll
- System.Text.Encoding.Extensions.dll
- System.Text.RegularExpressions.dll
- System.Threading.AccessControl.dll
- System.Threading.dll
- System.Threading.Overlapped.dll
- System.Threading.Tasks.dll
- System.Threading.Tasks.Parallel.dll
- System.Threading.Thread.dll
- System.Threading.ThreadPool.dll
- System.Threading.Timer.dll
- System.Transactions.dll
- System.ValueTuple.dll
- System.Web.Services.dll
- System.Windows.dll
- System.Xml.dll
- System.Xml.Linq.dll
- System.Xml.ReaderWriter.dll
- System.Xml.Serialization.dll
- System.Xml.XDocument.dll
- System.Xml.XmlDocument.dll
- System.Xml.XmlSerializer.dll
- System.Xml.XPath.dll
- System.Xml.XPath.XDocument.dll
- System.Xml.XPath.XmlDocument.dll
- System.Xml.Xsl.Primitives.dll
- Xamarin
- Xamarin.Android.NUnitLite.dll
- Xamarin.iOS.dll

- Xamarin.Mac.dll
- Xamarin.Mac.registrar.full.a
- Xamarin.Mac.registrar.mobile.a
- Xamarin.TVOS.dll
- Xamarin.TVOS.registrar.a
- Xamarin.WatchOS.dll
- Xamarin.WatchOS.registrar.a
- Xamarin-debug
- XamMac.CFNetwork.dll
- XamMac.dll
- XamMacLauncher

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